Complete Saturation of the Natural Operator Space $O\ell_p$

A candidate affirmative solution to Question 4.13 of arXiv:1704.07760

Automated mathematical discovery run `fa_banach_001`, lane 15

11 August 2026

Status. Candidate full solution, likely valid, requiring expert review. A bounded search found no explicit later resolution.

Abstract

We answer Question 4.13 of Corrêa affirmatively. In fact, for every $1 \le p < \infty$, including $p = 2$, the natural operator space $O\ell_p$ is $O\ell_p$-completely saturated. Every infinite-dimensional subspace of ℓ_p contains a normalized sequence summably close to a normalized disjoint block sequence. Disjoint normalized sequences are completely isometric to the canonical basis in the natural operator L_p structure, and a summable perturbation is small in completely bounded norm because it is a sum of rank-one maps. Choosing the total error below one completes the argument.

1 Source question

Corrêa equips ℓ_p with its natural operator-space structure $O\ell_p = (\min \ell_\infty, \max \ell_1)_{1/p}$ and asks whether it is $O\ell_p$-completely saturated [1].

Recall that an operator space E is E-completely saturated when every closed infinite-dimensional subspace contains a subspace completely isomorphic to E.

2 Two lemmas

We use the standard natural operator-space structure on the commutative L_p-space ℓ_p. Its complete Fubini identity is

$$S_p^n[O\ell_p] \cong \ell_p(S_p^n) \tag{1}$$

isometrically. A map between operator spaces is a complete isometry if and only if all its S_p^n-amplifications are isometries [2].

Lemma 1. *Let (u_k) be a disjointly supported sequence in ℓ_p with $\|u_k\|_p = 1$. The map $V : O\ell_p \to O\ell_p$ defined by $V(e_k) = u_k$ is a complete isometry onto its range.*

Proof. For a finite family $A_k \in S_p^n$, (1) and disjointness give

$$\begin{aligned}\left\|\sum_k A_k \otimes u_k\right\|_{S_p^n[O\ell_p]}^p &= \sum_j \left\|\sum_k u_k(j) A_k\right\|_{S_p^n}^p \\ &= \sum_k \sum_{j \in \operatorname{supp} u_k} |u_k(j)|^p \|A_k\|_{S_p^n}^p = \sum_k \|A_k\|_{S_p^n}^p.\end{aligned}$$

We do not know if Proposition 4.12 is true for $p = 1$.

One may wonder why the proof that being ℓ_p-saturated does not pass to the quantum case. For the proof presented in [8], the breaking point is the following: let N and M be closed subspaces of a Banach space X. If $N + M$ is closed, then we have the isomorphic identification:

$$\frac{N+M}{N} \cong \frac{M}{M \cap N}$$

If we let X be as in Proposition 4.8 and $Y = \{(0, y) \in do\ell_p(\theta) : y \in o\ell_p(\theta)\}$, then $(X + Y)/Y = o\ell_p(\theta)$ completely isomorphically, but Proposition 4.8 tells us that $X = X/(X \cap Y)$ is not completely isomorphic to $o\ell_p(\theta)$. That is, the second isomorphism theorem does not need to hold for operator spaces, even when it holds in the Banach setting (which is probably already known).

Recall that the space ℓ_p has a natural operator space structure, $O\ell_p$ (Section 3.2).

Question 4.13. Are the spaces $O\ell_p$, with their natural operator space structure, $O\ell_p-$completely saturated, $1 \leq p < \infty$, $p \neq 2$?

Question 4.14. Is being $O\ell_p-$completely saturated a C3SP?

4.2. **The isometric Palais' problem for operator spaces.** Having solved the isomorphic version of Palais' problem for operator spaces, it is easy to obtain a solution to the isometric version. We thank Gilles Pisier for pointing out how one can obtain a complete extension of OH which is isometric to a Hilbert space.

Figure 1: Question 4.13 and its context in the source, PDF p. 29.

The last expression is the norm of $\sum_k A_k \otimes e_k$ to the p-th power. The amplification criterion proves the claim. $\square$

Lemma 2. *If $(d_k) \subset O\ell_p$ and $\sum_k \|d_k\|_p = \eta < \infty$, then $D(e_k) = d_k$ extends to a completely bounded map on $O\ell_p$ with $\|D\|_{cb} \leq \eta$. Consequently, if V is a complete isometry and $\eta < 1$, then $V + D$ is a complete isomorphism onto a closed range.*

Proof. Let e_k^* be the k-th coordinate functional. Every bounded scalar-valued functional on an operator space has the same norm and cb norm, and the map $\lambda \mapsto \lambda d_k$ has cb norm $\|d_k\|_p$. Thus

$$D = \sum_k e_k^* \otimes d_k$$

converges absolutely in cb norm and $\|D\|_{cb} \leq \eta$. At every matrix level,

$$(1 - \eta)\|z\| \leq \|(V + D)_n z\| \leq (1 + \eta)\|z\|.$$

This gives a complete isomorphism and a closed range. $\square$

3 Affirmative solution

Theorem 1. *For every $1 \leq p < \infty$, the natural operator space $O\ell_p$ is $O\ell_p$-completely saturated. In particular, Question 4.13 of [1] has an affirmative answer.*

Proof. Let W be a closed infinite-dimensional subspace of ℓ_p. Fix positive numbers (δ_k) such that $2\sum_k \delta_k < 1$. We inductively construct integers

$$0 = N_0 < N_1 < N_2 < \cdots$$

and unit vectors $x_k \in W$. The restriction to W of the projection onto the first N_{k-1} coordinates has finite rank, so its kernel is infinite-dimensional. Choose x_k in that kernel with $\|x_k\|_p = 1$, and then choose N_k so that

$$\left\|x_k - P_{(N_{k-1},N_k]}x_k\right\|_p < \delta_k.$$

Put $v_k = P_{(N_{k-1},N_k]}x_k$ and $u_k = v_k/\|v_k\|_p$. The vectors u_k are normalized and disjointly supported. Moreover,

$$\|x_k - u_k\|_p \leq \|x_k - v_k\|_p + |1 - \|v_k\|_p| < 2\delta_k.$$

By Lemma 1, $V(e_k) = u_k$ is a complete isometry. By Lemma 2, the map $D(e_k) = x_k - u_k$ satisfies

$$\|D\|_{cb} \leq \sum_k \|x_k - u_k\|_p < 2\sum_k \delta_k < 1.$$

Hence $U = V + D$ is a complete isomorphism from $O\ell_p$ onto the closed span of (x_k). Since every x_k belongs to W, this closed span is contained in W. The arbitrary infinite-dimensional subspace W therefore contains a completely isomorphic copy of $O\ell_p$. $\square$

4 Verification and novelty bounds

The proof audit checks the gliding-hump construction, the factor two caused by normalizing the block vectors, the complete Fubini calculation, absolute cb-norm convergence of the rank-one perturbations, and the matrix-level lower bound for $V + D$. No finite computation is relevant.

The conclusion addresses complete saturation only. It does not answer the source's subsequent Question 4.14 asking whether this property is a complete three-space property.

A bounded search on 11 August 2026 covered the run's registry, solution, attempt, and active-claim indexes; the exact arXiv id and title; exact phrases from Question 4.13; combinations of $O\ell_p$, operator spaces, and complete saturation; published-paper metadata; and later citations of the source. No explicit answer was located. This is provisional novelty evidence only.

Human-review recommendation

Check first the complete Fubini identity (1) and the standard S_p amplification criterion. Then verify that the coordinate rank-one series converges in cb norm. The gliding-hump part is the usual elementary coordinate argument in ℓ_p and includes $p = 1$ without a weak-nullness assumption.

References

[1] W. H. G. Corrêa, *Twisting Operator Spaces*, Trans. Amer. Math. Soc. **370** (2018), 8921–8957; arXiv:1704.07760, Question 4.13.

[2] G. Pisier, *Introduction to Operator Space Theory*, Cambridge University Press, 2003; see the natural operator L_p structure, complete Fubini, and the S_p amplification criterion.